

Gujarat University
Department of Computer Science
Master of Computer Applications (MCA) – Semester 3
Subject: Computer Vision

Theory Questions

Questions

1. Explain image representation with reference to discrete and continuous image models. How does sampling and quantization convert a continuous image into a discrete digital image?
2. What is image enhancement? Explain the difference between spatial domain and frequency domain enhancement techniques with suitable examples.
3. Describe the process of image restoration. How does it differ from image enhancement? Explain any one degradation model used in restoration.
4. What is histogram equalization? Explain the mathematical procedure for performing histogram equalization on a grayscale image.
5. Explain histogram matching (histogram specification). How is it different from histogram equalization? Give an application where histogram matching is preferred.
6. Compare and contrast contour-based and region-based shape representation techniques. Give suitable examples for each.
7. What is contour-based representation? Explain chain codes and boundary descriptors used to represent object contours.
8. Explain region-based shape representation. Discuss the role of moments and region descriptors in representing a shape.
9. What are deformable curves and surfaces? Explain their significance in shape modelling and object boundary extraction.
10. Explain the concept of snakes (active contour models). Describe the energy minimization framework used in snakes for boundary detection.
11. Discuss the internal and external energy terms used in active contour models. How do they influence the evolution of the contour?
12. What are the limitations of traditional snake models? Explain how these limitations are addressed in improved active contour approaches.
13. Explain the level set method for shape and contour representation. How does it handle topological changes such as splitting and merging of contours?
14. Compare active contour (snake) models with level set representations. Discuss the advantages of level sets over parametric active contours.
15. What are Fourier descriptors? Explain how the Fourier transform of a shape boundary is used to represent and analyze the shape.

16. Explain wavelet descriptors for shape representation. How do wavelet-based methods differ from Fourier-based methods in capturing shape information?
17. What are medial representations of a shape? Explain the concept of the medial axis transform and its use in shape description.
18. Explain the concept of multiresolution analysis in image processing. How does it help in analyzing images at different scales?
19. Describe the role of the Gaussian and Laplacian pyramids in multiresolution image analysis.
20. Compare Fourier descriptors, wavelet descriptors, and medial axis representation as shape description techniques, highlighting their advantages and limitations.